

Joshua Kennedy

Cloud Platform Engineer
Resides in Scotland

Experience

Cloud Platform Engineer

NatWest - October 2023 - Present

- > Engineered strategic on-premise Container-as-a-Service (CaaS) capabilities supporting 32 OpenShift clusters across four environments for approximately 300 engineering teams.
- > Developed platform engineering capabilities through policy-driven automation and GitLab CI/CD, delivering centralised GitOps orchestration, HashiCorp Vault integration and customer guardrails.
- > Designed reusable platform capabilities consumed by approximately 300 application teams, reducing operational complexity while improving consistency and self-service.
- > Collaborated within a 12-person platform engineering team alongside principal engineers, architects and stakeholders to deliver shared platform capabilities in a regulated financial services environment.
- > Continuously evaluated, prototyped and implemented improvements to platform tooling, automation and operational workflows to increase platform maturity and developer productivity.

Cloud Automation Engineer

NatWest - September 2021 - October 2023

- > Joined a small R&D team focused on expanding the bank's on-premise cloud computing capabilities.
- > Built experience across OpenShift, Kubernetes, container orchestration, Linux and cloud platform automation.
- > Delivered proof-of-concept work for certificate automation using cert-manager and centralised secret management using HashiCorp Vault.
- > Collaborated across teams to turn emerging platform requirements into working prototypes and executable patterns.
- > Supported automation and CI/CD-aligned delivery practices to reduce manual effort and increase consistency across platform operations.

Middleware Systems Engineer

NatWest - February 2019 - September 2021

- > Built automation to streamline common support tasks for deployment orchestration tooling within a feature team.
- > Automated agent upgrade workflows, enabling over 1,600 agents to be upgraded across approximately 50 teams within six months.
- > Supported users and administrators of deployment orchestration platforms, producing documentation and running workshops to upskill a wider support team.
- > Improved platform support processes by combining scripting, documentation, incident analysis and close collaboration with consuming teams.

Middleware Technical Analyst

RBS - January 2018 - February 2019

- > Delivered technical support for Java-based enterprise middleware, resolving over 150 individual incidents.
- > Developed infrastructure support scripts and helped stand up deployment orchestration tooling.
- > Built practical experience with REST API integration, high-availability clustering and production support in a regulated banking environment.

WebSphere Technical Apprentice

RBS - January 2016 - January 2018

- > Developed KornShell automation to replace a manual licensing process across approximately 5,000 servers.
- > Audited and improved certificate renewal tracking workflows, building skills in Visual Basic, SQL and enterprise middleware administration.
- > Completed Modern and Graduate Apprenticeship pathways while working in production financial services technology teams.

Personal Projects

tapo-onboarding

Shell / Reverse Engineering / IoT

Reverse engineered TP-Link's encrypted provisioning protocol to create one of the first open-source toolchains capable of onboarding modern Tapo cameras without vendor cloud services, solving a problem previously lacking a practical community solution.

Stateful

JavaScript / Pebble / REST APIs

Developed a generic REST client for Pebble that enabled arbitrary REST APIs to be modelled and controlled from the watch without assumptions about the underlying service, addressing a capability absent from existing applications on the platform.

restate-go

Go / Home Automation / APIs

Developing a local-first home automation platform that prioritises ownership, privacy and reliability over cloud-dependent architectures, exploring an alternative to conventional commercial ecosystems.

Snowdon II WiFi

C++ / Raspberry Pi Pico / REST APIs / Infrared

Engineered an embedded hardware modification that adds Wi-Fi and a REST API to a consumer soundbar through infrared protocol emulation and state inference, creating capabilities unavailable from the manufacturer.

Education and Certification

Certificate of Higher Education in Engineering

April 2023 - October 2025

Red Hat OpenShift Administration II: Configuring a Production Cluster (DO280)

October 2024

Red Hat Certified Specialist in Containers and Kubernetes

July 2022

Graduate Apprenticeship Scotland in IT & Telecommunications Professionals

January 2017 - November 2017

Modern Apprenticeship Scotland in IT & Telecommunications Professionals

January 2016 - November 2016

HND Computer Games Development

August 2013 - August 2015

Technical Skills

Platform Engineering

Software Engineering

OpenShift / Kubernetes

Automation

CI/CD

Linux

Infrastructure as Code

Go

Shell / Bash / KornShell

REST APIs

Python

JavaScript

Embedded C / C++

Security & Policy Controls

Professional Skills

Technical Leadership

Stakeholder Communication

Cross-functional Collaboration

Problem Solving

Continuous Improvement

Mentoring

Contact

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